

Marika Nishi

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EDUCATION

University of Pennsylvania, School of Engineering & Applied Science Philadelphia, PA, United States	May 2027
Candidate for Master of Science in Engineering, <i>Majors: Robotics, Computer Science</i>	GPA: 3.81/4.0
University of Tokyo, Faculty of Engineering Bunkyo, Tokyo, Japan	March 2023
Bachelor of Engineering, <i>Major: Mechanical Engineering</i>	GPA: 3.64/4.0

SKILLS & RELEVANT COURSEWORK

Software Skills: Python, PyTorch, C++, C, ROS, ROS2, ML, AI, perception, computer vision, SLAM, localization, sensor fusion, EKF, OpenCV, motion planning, ML-based control, simulation (Gazebo, Isaac Sim), autonomous driving, drones
Hardware Skills: CAN, LiDAR, RTK-GNSS, GNSS, IMU, RGB cameras, depth sensors, motors, microcontroller, CAD
Relevant Coursework: Mechatronics; F1tenth Autonomous Racing Cars; Automotive Engineering; Deep Neural Networks; Deep Learning; Computer Vision; Dynamics and Control; Machine Perception; Machine Learning
Publication: Hori, K., Watanabe, K., Tanaka, T., **Nishi, M.**, & Ito, T. (2026). Visibility Evaluation Around Unsignalized Intersections with Occlusion From Arbitrary On-road Viewpoints Using Point Cloud Maps. *International Journal of Intelligent Transportation Systems Research*.
Watanabe, K., **Nishi, M.**, & Ito, T. (2025). Placement design of various roadside sensors for stochastic position prediction of traffic participants on community roads. *International Journal of Intelligent Transportation Systems Research*.

EMPLOYMENT EXPERIENCE

Vijay Kumar Lab <i>Research Assistant</i> , Philadelphia, PA, United States	May 2026 – Present
Design and build VLA control system of drones with robotic arms to complete tasks like turning valves with IsaacSim	
ISUZU <i>Advanced Engineering - AI & Autonomous Driving Intern</i> , Plymouth, MI, United States	May 2026 – August 2026
- Investigated autonomous truck on-road test failures by tracing issues across a large-scale perception, planning, and control software architecture built with Bazel and protobuf, identifying root causes through code analysis and debugging - Used SQL to analyze autonomous driving on-road test data and quantitatively evaluate performance on dashboards	
Vijay Kumar Lab <i>Research Intern</i> , Philadelphia, PA, United States	May 2025 – May 2026
- Implemented driving simulators that visualized data of event camera on F1tenth car using Gazebo, CARLA, and ROS - Collected datasets on driving simulator and trained vision transformer model that computed appropriate motion control from event camera images; improved data generation process efficiency by removing redundancy, cutting time by 70 %	
BOSCH <i>Software Intern</i> , Tsuzuki, Kanagawa, Japan	August 2023 – September 2023
- Designed and built map to enhance autonomous driving by visualizing key information, such as routes and vehicle state using MATLAB; helped company's engineers understand signals sent from cars visually - Improved software that processed CAN signals from cars by fixing signal decoding; refined system output accuracy - Collaborated with colleagues from 10 countries and different majors; made outcomes reflecting diverse opinions	
DMG MORI <i>Industrial Practices Intern, Additive Manufacturing R & D</i> , Iga, Mie, Japan	August 2021 – September 2021
- Analyzed performance tests of additive manufacturing machines by operating the machines and inspecting products - Investigated additive manufacturing methods by examining machine structures; proposed solutions to heat distortion	

ROBOT AUTOMATION PROJECTS

F1tenth Autonomous Racing Cars <i>Perception, Planning, Control, Autonomous Driving, Python, C++, ROS2</i>	January 2025 – May 2025
Developed a 1/10-scale autonomous racing car; implemented SLAM localization and supported integration and testing of motion planning and vehicle control systems using Python, C++, and ROS; won second place for fastest lap in race	
Research of Traffic Collision Prediction System <i>Prototyping, Extended Kalman Filter, C++, ROS</i>	April 2022 – March 2023
- Designed and built real-time collision prediction system among pedestrian, cyclist, and intelligent wheelchair - Wrote C++ ROS codes that detected pedestrians and cyclists based on LiDAR data - Conducted sensor fusion using Extended Kalman Filter; estimated pedestrians and cyclists' positions - Operated RTK-GPS, low-accuracy GPS, LiDAR, and IMU to collect and integrate data into system development	